Method for Walk Correction using Leading Edge Slope:

Derived from dual threshold feature of TOFHIR2C

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Abstract

Precision timing at the HL-LHC requires rapid, repeatable calibration as detector response evolves. With many interactions per bunch crossing, an external particle-arrival time reference is not readily available. The CMS barrel timing layer (BTL) therefore motivates calibration procedures based on internal measurements and a consistent parametrization across about 330 000 channels.

We extend an earlier dual-threshold bench study to BTL sensors and TOFHIR2C readout exposed to a 180 GeV pion beam. Two threshold-crossing times provide a leading-edge slope proxy. Its mean dependence on charge defines an inverse-slope walk correction with one amplitude parameter, AWC, per channel and configuration.

Comparison with independent microchannel-plate (MCP) reference calibrations shows good agreement between the correction shapes. Internal charge and slope information provide candidate predictors of AWC, with the MCP results serving as benchmarks. These results support development of a reference-free calibration procedure; predictive precision and transfer to other operating conditions require further validation.

BTL and TOFHIR2C: Refs. [3, 7]; dual-threshold motivation and bench study: Ref. [5].

Charge-based calibration works, but its coefficients vary

The MCP reference note obtains good walk descriptions with a quadratic in inverse charge [6]:

$$W_i(Q) = a_i + b_i/Q + c_i/Q^2.$$

Bar	SiPM	a_i [ns]	b_i [ns ADC]	c_i [ns ADC^2]
7	Left	0.0796	238.3	-2.156×10^4
7	Right	-0.5155	422.6	-5.583×10^4
8	Left	0.1218	198.5	-1.523×10^4
8	Right	-0.2292	388.9	-5.132×10^4

With Q expressed as raw, uncalibrated ADC values, the shape coefficients vary substantially across these four SiPMs.

Two bars in the reference module. Earlier time-of-flight calibration experience: ALICE [1, 2]. Q is not gain-calibrated across SiPMs; a_i includes an arbitrary timing offset. Coefficient variation alone is not a measure of calibration precision.

TOFHIR2C measures the leading-edge slope with two thresholds

Larger pulses cross a fixed threshold earlier: the timestamp depends on amplitude [4]. Two-threshold timing has earlier implementations [8, 9]; TOFHIR2C provides both crossings [7, 5].

$$I_1 < I_2, \qquad t_1 < t_2, \qquad S = \frac{I_2 - I_1}{t_2 - t_1} \quad [\mu\mathrm{A/ns}].$$

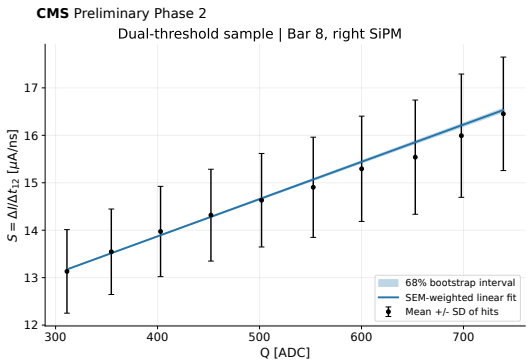
Locally, an approximately linear leading edge gives

$$\Delta t \simeq \frac{\Delta I}{S}.$$

Physical model: describe amplitude walk with an inverse slope, rather than an independent polynomial shape for every channel.

S is a finite-interval slope proxy, not the instantaneous derivative. The dual-threshold sample uses $\Delta I = 7.8125\,\mu\mathrm{A}$ from the implemented threshold conversion. The fitted AWC need not equal this threshold separation. GitHub: timing and threshold reconstruction [11].

Measure the charge-to-slope mapping using internal hits



Typical single-hit relative uncertainties: Direct slope: $\sim 15\%$; charge: $< 5\%$. Use the more precise Q to estimate the mean slope:

$$\overline{S}_i(Q) = c_{0,i} + c_{1,i} Q.$$

Ten SEM-weighted profile means; no MCP time.

Reference module, bar 8, right SiPM: 19 312 hits. Both SiPMs required; other module unconstrained. Physical threshold crossings; peak-right charge selection. Points: mean slopes; bars: hit SD, not mean uncertainty. Band: conditional 68% interval from 1000 hit-bootstrap refits with fixed bins.

One amplitude describes walk once the slope is calibrated

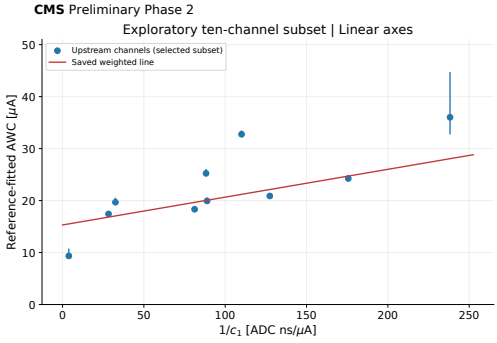
$$W(Q) = \frac{\mathrm{AWC}}{\overline{S}(Q)}, \qquad t_{\mathrm{corr}} = t_{\mathrm{measured}} - W(Q).$$

- Internal hits determine $\overline{S}(Q)$, the mean slope at a given charge, for each SiPM.
- One parameter, AWC (in μA), sets the walk magnitude.
- AWC is constant per SiPM and configuration; the correction varies with Q.
- Channel timing offsets are a separate calibration.

The remaining task is to predict AWC from internal observables. Use the MCP-calibrated channels to develop and test that prediction.

Apply the model within its calibrated charge range, with $\overline{S}(Q) > 0$. The slope fit alone does not fix AWC. The common functional form follows the dual-threshold model in Ref. [5].

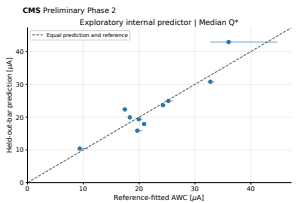
Candidate 1: AWC versus inverse slope coefficient



An exploratory trend: AWC versus $1/c_1$, the inverse slope-to-charge coefficient. Each point is one SiPM, not a charge bin. AWC is benchmarked against MCP; c_1 uses internal hits. The ten-SiPM trend motivates prediction, but the line is not a precision calibration ($\chi^2/\mathrm{ndf} = 538.6/8$). Bars: slope-only 16–84% bootstrap intervals; reference errors and correlations are omitted. More independent data are needed.

Candidate 2: a physics-motivated AWC predictor

Physics motivation: similar scintillation pulses and front-end shaping motivate testing a shared timing sensitivity to the same fractional charge change.



$$K \simeq -\left.\frac{dW}{d\ln Q}\right|_{Q_*}, \quad \mathrm{AWC}_{\mathrm{pred}} = \frac{K\overline{S}(Q_*)^2}{c_1 Q_*}.$$

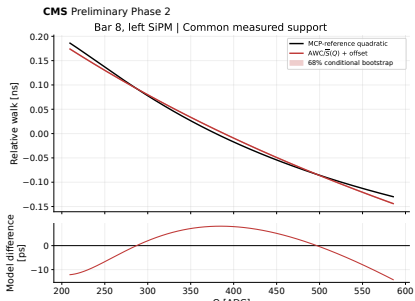
Required assumptions

1. $W = \mathrm{AWC}/\overline{S}$ holds locally, with constant AWC per SiPM.
2. The fitted $\overline{S}(Q) = c_0 + c_1 Q$ is reliable near Q_*.
3. Q_*, the selected-charge median per SiPM, samples comparable pulse regions.
4. Similar pulses and operating conditions allow a common K across SiPMs and bars.

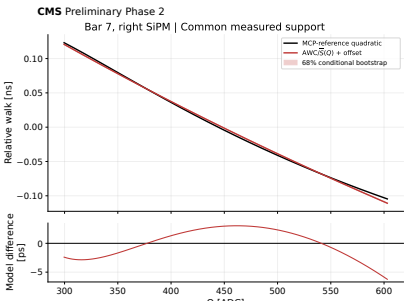
Ten SiPMs; hold out each of six bars and learn K from the others' MCP-calibrated AWC. RMS error: 3.22 μA; fractional RMS: 14.1%. Dashed line: equality. Horizontal bars: conditional reference-AWC intervals, not prediction intervals. Not a universal law: full uncertainties, correlations and transfer to new configurations remain unvalidated.

The single-amplitude model closely follows the MCP curves

Bar 8, left SiPM

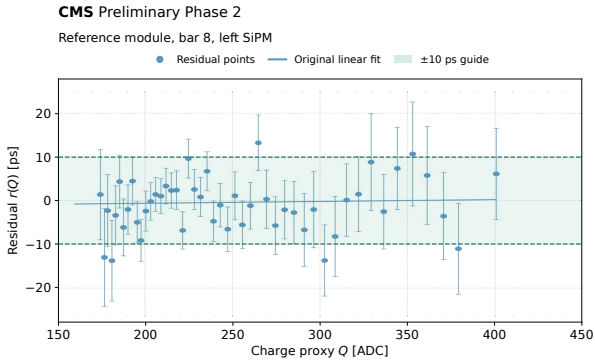


Bar 7, right SiPM



Reference module. Black: MCP quadratic; red: dual-threshold model. Different-event samples; AWC and offset fitted only in common populated charge intervals. Bar 8, left SiPM: 6.80 ps RMS curve mismatch. Lower panels show curve differences, not event resolution. Bands propagate slope-fit statistics with the MCP curve fixed. MCP clock alignment uses LOWESS [10]. This benchmarks shapes, not AWC prediction.

The AWC correction describes the walk at the 10 ps scale



- **Y axis:** mean time minus MCP, the AWC correction and a constant offset.
- **Near-zero, flat trend:** little charge-dependent timing bias remains.
- ± 10 **ps guide:** supports the AWC walk description at this scale, not an all-point bound.

Residual definition: $r(Q) = \langle t_{\mathrm{thr1}} - t_{\mathrm{MCP}} \rangle_Q - \mathrm{AWC}/\overline{S}(Q) - b$, where b removes a constant timing offset. Original Figure 5 and MCP-reference data [6]: points, error bars and linear fit retained; ns converted to ps. The guide is not an uncertainty band or single-hit resolution; this tests shape, not reference-free AWC prediction.

References

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[7] E. Albuquerque et al., "TOFHIR2: The readout ASIC of the CMS Barrel MIP Timing Detector", arxiv:2404.01208

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[10] W. S. Cleveland, "Robust Locally Weighted Regression and Regression Scatterplots", J. Amer. Statist. Assoc. **74** (1979) 829.

[11] GitHub repository, https://github.com/De-Cristo/LecroyTranslator4CMSTimingDetector.